

DILLON, MT, USA

WMO: 727700

Lat: 45.258N

Lon: 112.554W

Elev: 1585

StdP: 83.68

Time Zone: -7.00 (NAM)

Period: 97-19

WBAN: 24138

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-23.0	-18.9	-26.6	0.4	-22.8	-22.6	0.6	-18.4	12.5	-0.5	11.4	0.2	3.0	50	0.669

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	31.4	15.2	29.6	14.8	27.9	14.2	16.9	27.6	16.0	26.6	15.2	25.4	4.1	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.1	11.4	20.1	12.2	10.7	19.4	11.1	10.0	18.7	53.3	27.6	50.2	26.6	47.5	25.4	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.8	10.4	8.9	-27.5	34.4	4.1	1.2	-30.4	35.2	-32.8	35.9	-35.1	36.6	-38.0	37.5		
			-27.7	19.8	4.0	2.2	-30.6	21.3	-33.0	22.6	-35.2	23.8	-38.1	25.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	-4.2	-3.5	1.4	5.3	10.1	14.4	19.5	18.2	13.5	6.6	0.0	-5.5	(6)
	DBStd	9.79	5.80	6.09	5.43	4.29	4.33	3.91	2.67	2.87	4.39	5.11	6.41	6.07	(7)
	HDD10.0	2221	442	377	267	151	53	9	0	0	18	126	299	480	(8)
	HDD18.3	4467	700	611	524	392	256	128	18	36	153	363	549	738	(9)
	CDD10.0	900	0	0	1	9	57	139	296	256	121	21	1	0	(10)
	CDD18.3	104	0	0	0	0	1	8	56	33	6	0	0	0	(11)
	CDH23.3	2023	0	0	0	1	47	212	910	661	185	7	0	0	(12)
	CDH26.7	648	0	0	0	0	6	57	328	215	41	1	0	0	(13)
(14) Wind	WSAvg	4.0	4.6	4.4	4.5	4.3	4.0	3.6	3.3	3.4	3.6	3.9	4.2	4.1	(14)
(15) Precipitation	PrecAvg	291	9	6	15	32	53	51	29	30	26	19	11	8	(15)
	PrecMax	487	26	23	45	78	124	125	108	83	84	57	36	30	(16)
	PrecMin	146	0	0	1	2	6	7	0	0	0	0	0	0	(17)
	PrecStd	75	8	6	10	19	29	28	24	19	20	16	8	7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.1	11.5	18.0	22.4	27.4	31.2	33.8	32.8	30.1	24.0	17.1	8.5	(19)
		MCWB	3.1	3.9	6.7	9.2	12.2	14.8	16.6	15.0	13.8	10.8	7.3	3.3	(20)
	2%	DB	6.3	8.5	14.8	19.6	24.3	28.0	32.0	31.1	27.8	20.8	13.3	6.2	(21)
		MCWB	1.8	2.7	5.4	8.0	11.3	14.3	15.9	14.7	13.0	9.5	5.9	2.0	(22)
	5%	DB	4.4	6.3	12.2	16.7	22.1	25.9	30.1	29.0	25.8	17.9	10.9	4.1	(23)
		MCWB	0.6	1.4	4.5	6.8	10.7	13.8	15.5	14.1	12.1	8.2	4.6	0.6	(24)
	10%	DB	2.8	4.2	9.7	13.8	19.0	23.1	28.3	27.4	22.9	15.2	8.2	2.4	(25)
		MCWB	-0.4	0.1	3.2	5.5	9.5	12.7	15.0	13.7	11.4	7.3	3.2	-0.6	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.0	4.6	7.6	10.1	14.0	17.3	19.1	17.4	15.8	11.9	8.3	3.9	(27)
		MCDB	7.9	10.4	16.4	20.5	23.1	27.7	30.2	29.2	25.2	21.6	14.8	7.9	(28)
	2%	WB	2.2	3.0	6.0	8.5	12.4	15.6	17.5	15.9	14.2	10.2	6.3	2.4	(29)
		MCDB	5.9	7.6	13.7	17.8	21.7	25.6	27.9	26.7	24.6	18.9	12.7	5.8	(30)
	5%	WB	0.9	1.7	4.8	7.2	11.2	14.3	16.5	15.2	13.0	8.9	4.9	0.9	(31)
		MCDB	4.2	6.0	11.3	15.6	20.1	23.4	27.2	25.8	22.8	16.6	10.2	3.9	(32)
	10%	WB	-0.1	0.5	3.6	6.0	10.1	13.3	15.7	14.4	11.9	7.6	3.5	-0.4	(33)
		MCDB	2.8	4.0	9.1	13.1	18.2	22.0	26.2	25.0	21.3	14.6	8.0	2.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	10.4	11.8	12.8	13.5	14.7	17.8	17.7	15.6	13.2	10.8	9.3	(35)
		MCDBR	10.1	11.7	14.7	17.6	18.0	18.9	20.2	20.4	19.7	17.0	13.3	9.7	(36)
	5% WB	MCWBR	6.9	7.4	7.9	8.4	7.9	8.3	8.6	8.6	8.6	8.7	7.5	6.8	(37)
		MCWBR	9.5	10.8	13.7	16.3	16.3	17.4	18.3	18.2	17.3	15.8	12.5	9.5	(38)
(40) Clear-Sky Solar Irradiance	taub		0.236	0.244	0.262	0.297	0.309	0.318	0.327	0.328	0.300	0.273	0.252	0.235	(40)
		taud	2.411	2.402	2.493	2.450	2.492	2.500	2.497	2.461	2.539	2.587	2.547	2.470	(41)
	Ebn at Noon		905	961	985	966	959	947	934	922	929	917	884	874	(42)
		Edh at Noon	69	86	91	105	104	104	103	103	87	72	61	59	(43)
(44) All-Sky Solar Radiation	RadAvg		1.62	2.60	3.89	5.07	5.81	6.74	7.16	6.17	4.70	3.01	1.83	1.32	(44)
	RadStd		0.15	0.22	0.24	0.37	0.54	0.54	0.37	0.34	0.39	0.24	0.14	0.10	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	-0.66	N/A	N/A	N/A	N/A

Nomenclature: See separate page